NEET TEST HUB

NTH-PREMIUM TEST SERIES 08

Date – 16 November 2025	Duration – 3 Hours	Marks – 720

Physics :- Oscillations, Waves

Chemistry :- Organic Chemistry: Some Basic Principles & Techniques, Hydrocarbons

Biology :- Cell-The Unit Of Life, Biomolecules, Cell Cycle & Cell Division

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. Given below are two statements:

Statement I:	A seconds pendulum has a time period of 1 second.
Statement II:	A seconds pendulum takes exactly 1 second to travel between its two extreme positions.

1.	Both **Statement I** and **Statement II** are incorrect.
2.	**Statement I** is incorrect and **Statement II** is correct.
3.	**Statement I** is correct and **Statement II** is incorrect.
4.	Both **Statement I** and **Statement II** are correct.

2. Given below are two statements:

Assertion (A):	In a progressive wave motion, the particle velocity remains constant at all times.
Reason (R):	During the propagation of sound, the density of the medium is uniform throughout.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

3. Two simple pendulums of length 1 m and 1.44 m are at their mean position with their velocities in the same direction at some instant. After how many oscillations of bigger pendulum they will again be in the same phase?

1.	5	2.	6
3.	7	4.	8

4. A travelling wave pulse is given by the equation;

$$y = \frac{4}{3x^2 + 48t^2 + 24xt + 2},$$

where x and y are in metres and t is in seconds. The velocity of the wave is:

1. $4\ \mathrm{m/s}$
2. $2\ \mathrm{m/s}$
3. $8\ \mathrm{m/s}$
4. $12\ \mathrm{m/s}$

5. In a stationary wave:

1.	strain is minimum at the nodes
2.	strain is maximum at the nodes
3.	strain is maximum at the antinodes
4.	amplitude is zero at all points.

6. A wall clock uses a vertical spring-mass system to measure the time. Each time the mass reaches an extreme position, the clock advances by a second. The clock gives the correct time at the equator. If the clock is taken to the poles it will:

1.	run slow	2.	run fast
3.	stop working	4.	give correct time

7. A simple pendulum oscillating in air has a period of $\sqrt{3}$ s. If it is completely immersed in non-viscous liquid, having density $\left(\frac{1}{4}\right)^{\text{th}}$ of the material of the bob, the new period will be:

1.	$2\sqrt{3}$ s	2.	$\frac{2}{\sqrt{3}}$ s
3.	2 s	4.	$\frac{\sqrt{3}}{2}$ s

8. A sound wave with a frequency of 100 kHz is propagating through the air and encounters a water surface. The transmitted sound wave in the water will have:

1.	the same speed but a different frequency
2.	a different speed and a different frequency
3.	the same speed but a different wavelength
4.	a different speed and a different wavelength

9. A particle moves in the x-y plane according to the equation

$x = A\cos^2 \omega t$ and $y = A\sin^2 \omega t$

Then, the particle undergoes:

1.	uniform motion along the line $x + y = A$
2.	uniform circular motion along $x^2 + y^2 = A^2$
3.	SHM along the line $x + y = A$
4.	SHM along the circle $x^2 + y^2 = A^2$

10. The differential equation representing the simple harmonic motion of a particle is $16\dfrac{d^2y}{dt^2} + 9y = 0$. If the particle starts at the mean position, the time taken for the particle to reach half of its amplitude for the first time will be:

1.	$\dfrac{4\pi}{9}$ s	2.	$\dfrac{5\pi}{9}$ s
3.	$\dfrac{\pi}{3}$ s	4.	$\dfrac{2\pi}{9}$ s

11. The given, equation describes a wave:

$y = A\sin 314\left[\dfrac{t}{0.5\text{ s}} - \dfrac{x}{100\text{ m}}\right]$.

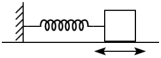
Identify the correct relationships regarding its frequency n and wavelength λ.

(A) $n = 2$ Hz
(B) $n = 100$ Hz
(C) $\lambda = 2$ m
(D) $\lambda = 100$ m

Choose the correct option from the given ones:

1.	**(A), (B)** and **(C)** only
2.	**(A)** and **(B)** only
3.	**(B)** and **(C)** only
4.	**(B), (C)** and **(D)** only

12. A block, when suspended from a spring, causes it to extend by a length l, in equilibrium. If the block is attached to the same spring and allowed to oscillate the time period of oscillation will be:



1.	$2\pi\sqrt{\dfrac{l}{g}}$	2.	$\pi\sqrt{\dfrac{l}{g}}$
3.	$2\sqrt{\dfrac{l}{g}}$	4.	$2\sqrt{\dfrac{g}{l}}$

13. The time taken by a particle performing SHM to pass from the point A to B where its velocities are same is 2 s. After another 2 s, it returns to point B. The time period of oscillation is:

1.	2 s	2.	8 s
3.	6 s	4.	4 s

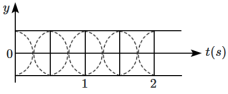
14. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is:

1.	2π s	2.	π s
3.	2 s	4.	1 s

15. A simple pendulum undergoing small oscillations of amplitude 1 cm, has a time period of 2 s. If the amplitude of oscillation is halved, the time period will:

1.	remain unchanged
2.	increase to 4 s
3.	decrease to 1 s
4.	decrease to 0.5 s

16. The correct figure that shows, schematically, the wave pattern produced by the superposition of two waves of frequencies 9 Hz and 11 Hz, is:

1.


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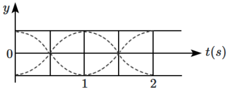

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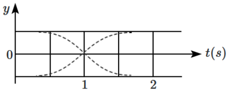

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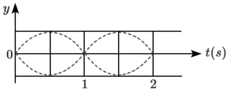


17. The displacement x of a particle varies with time t as $x = Asin\left(\frac{2\pi t}{T} + \frac{\pi}{3}\right)$. The time taken by the particle to reach from $x = \frac{A}{2}$ to $x = -\frac{A}{2}$ will be:

1.	$\frac{T}{2}$	2.	$\frac{T}{3}$
3.	$\frac{T}{12}$	4.	$\frac{T}{6}$

18. If the fundamental frequency of the string is 220 cps, the frequency of its fifth harmonic will be:

1.	44 cps	2.	55 cps
3.	1100 cps	4.	440 cps

19. A sonometer wire supports a 5 kg load and vibrates in fundamental mode with a tuning fork of frequency 420 Hz. The length of the wire between the bridges is now doubled. In order to maintain fundamental mode with same frequency, the load should be changed to:

1.	1 kg	2.	2 kg
3.	8 kg	4.	20 kg

20. Which, of the following, is true for a simple pendulum undergoing small oscillations?
(neglect all dissipative forces)

1.	Kinetic energy is conserved
2.	Momentum is conserved
3.	Potential energy is conserved
4.	Total energy is conserved

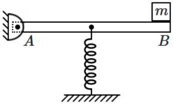
21. A wave is represented by the equation $y=A\sin\left(10\pi x+15\pi t+\frac{\pi}{3}\right)$ where x is in meters and t is in seconds. The expression represents:

1.	a wave traveling in the positive x-direction with a velocity 1.5 m/s
2.	a wave traveling in the negative x-direction with a velocity 2.5 m/s
3.	a wave traveling in the negative x-direction having a wavelength 0.2 m
4.	a wave traveling in the positive x-direction having a wavelength 0.2 m

22. When two tuning forks with nearly equal frequencies (say: 100 Hz and 102 Hz) are sounded together,

1.	Beats of frequency 2 Hz are heard
2.	Beats of frequency 101 Hz are heard
3.	Echo of frequency 202 Hz is heard
4.	Standing waves of frequency 202 Hz are formed

23. A light rod AB is hinged at A so that it is free to rotate about A. It is initially horizontal with a small block of mass m attached at B, and a spring (constant - k) holding it vertically up at its mid-point. The time period of vertical oscillations of the system is:



1.	$2\pi\sqrt{\frac{m}{k}}$	2.	$\pi\sqrt{\frac{m}{k}}$
3.	$4\pi\sqrt{\frac{m}{k}}$	4.	$\frac{\pi}{2}\sqrt{\frac{m}{k}}$

24. If the angular frequency of the given motion $y = sin(\omega t) + cos(\omega t)$ is $k\omega$, then value of k is:
1. 1/2
2. 1
3. 2
4. none of these

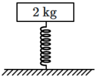
25. The pressure wave $P = 0.01 \sin(1000t - 3x)\ \text{Nm}^{-2}$, corresponds to the sound produced by a vibrating blade on a day when the atmospheric temperature is 0°C. On some other day, when the temperature is T, the speed of sound produced by the same blade and at the same frequency is found to be $336\ \text{ms}^{-1}$. The approximate value of T is:
1. 4°C
2. 12°C
3. 11°C
4. 15°C

26. In the sonometer experiment, a string of mass 18 g having linear mass density 20 g/m oscillates in the fundamental mode (of frequency 50 Hz). The velocity of transverse waves in the string is:
1. 70 m/s
2. 60 m/s
3. 90 m/s
4. 110 m/s

27. An air column in a pipe that is closed at one end will be in resonance with a vibrating tuning fork of frequency 330 Hz if the length of the column is:
(Speed of the sound in air is $330\ \text{m/s}$)
1. 50 cm
2. 75 cm
3. 150 cm
4. 100 cm

28. A uniform thin rope of length $12\ \text{m}$ and mass 6 kg hangs vertically from a rigid support and a block of mass 2 kg is attached to its free end. A transverse short wave-train of wavelength 6 cm is produced at the lower end of the rope. What is the wavelength of the wavetrain (in cm) when it reaches the top of the rope?
1. 12
2. 9
3. 3
4. 6

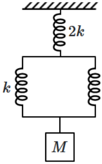
29. A platform with a mass of 2 kg is supported by a spring of negligible mass, as shown in the figure. The platform oscillates with a period of 3 s when it is pushed down and released. What must be the mass of a block that must be placed on the platform to double the period of oscillation to 6 s?



1.	1 kg	2.	2 kg
3.	4 kg	4.	6 kg

30. If two identical springs of stiffness k (each) are combined end-to-end, the resulting spring has a stiffness:
1. k
2. $2k$
3. $\frac{k}{2}$
4. zero

31. What is the period of oscillation of the block shown in the figure?

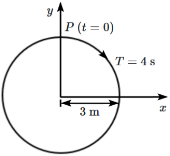


1.	$2\pi\sqrt{\frac{M}{k}}$	2.	$2\pi\sqrt{\frac{4M}{k}}$
3.	$\pi\sqrt{\frac{M}{k}}$	4.	$2\pi\sqrt{\frac{M}{2k}}$

32. For a particle performing linear SHM, its position (x) as a function of time (t) is given by $x = A\sin(\omega t + \delta)$. If, at $t = 0$, particle is at $+\frac{A}{2}$ and is moving towards $x = +A$, then $\delta =$

1.	$\frac{\pi}{3}$ rad	2.	$\frac{\pi}{6}$ rad
3.	$\frac{\pi}{4}$ rad	4.	$\frac{5\pi}{6}$ rad

33. The radius of the circle, the period of revolution, initial position and direction of revolution are indicated in the figure.

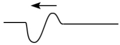
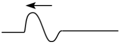
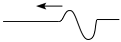
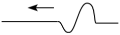


The y-projection of the radius vector of rotating particle P will be:

1. $y(t) = 3\cos\left(\frac{\pi t}{2}\right)$, where y in m
2. $y(t) = -3\cos 2\pi t$, where y in m
3. $y(t) = 4\sin\left(\frac{\pi t}{2}\right)$, where y in m
4. $y(t) = 3\cos\left(\frac{3\pi t}{2}\right)$, where y in m

34. For a simple pendulum, the graph between T^2 and L is:
1. a straight line passing through the origin
2. parabola
3. circle
4. ellipse

35. The wave shown in traveling to the right. Which of the waves below, traveling to the left, will momentarily cancel this wave?

1.		2.	
3.		4.	

36. n waves are produced on a string in one second. If the radius of the string is doubled and the tension is maintained the same, the number of waves produced in one second for the same harmonic will be:

1.	$2n$	2.	$\frac{n}{3}$
3.	$\frac{n}{2}$	4.	$\frac{n}{\sqrt{2}}$

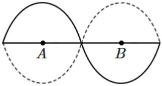
37. For the wave equation, $y = A\sin(Bt - Cx)$, match **Column I** with **Column II:**

	Column I		Column II
(a)	Wave speed	(p)	$\frac{B}{2\pi}$
(b)	Maximum particle speed	(q)	$\frac{C}{2\pi}$
(c)	Wave frequency	(r)	$\frac{B}{C}$
(d)	Wavelength	(s)	None of these

Codes:

1.	a - s, b - p, c - q, d - r
2.	a - r, b - p, c - q, d - s
3.	a - s, b - q, c - p, d - r
4.	a - r, b - s, c - p, d - s

38. What is the phase difference between the particles located at positions A and B in the standing wave depicted?



1.	0	2.	$\frac{\pi}{2}$
3.	$\frac{5\pi}{6}$	4.	π

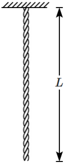
39. A simple pendulum has some time period T. What will be the percentage change in its time period if its amplitude is decreased by 5%?

1.	6%
2.	3%
3.	1.5%
4.	it will remain unchanged

40. A heavy uniform rope is suspended from the ceiling, and a block equal to 3 times the mass of the rope is suspended from the bottom. The ratio of the speed of a transverse pulse at the top (v_t) to its speed at the bottom (v_b) of the rope, $\frac{v_t}{v_b} =$

1. $\frac{4}{3}$
2. $\sqrt{\frac{4}{3}}$
3. $\frac{3}{4}$
4. $\sqrt{\frac{3}{4}}$

41. A heavy uniform rope hanging under its own weight from the ceiling of a room, can propagate small disturbances along its length as it is under tension. Let the mass per unit length of the rope be μ, and its total length L.

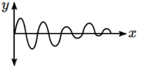
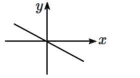
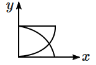


The speed of a wave-like transverse disturbance along the rope is given by $\sqrt{\frac{T}{\mu}}$, where T is the tension at the location. The speed of waves, at a distance x from the bottom, is:

1.	$\sqrt{g(L-x)}$	2.	$\sqrt{gx}$
3.	$\sqrt{2g(L-x)}$	4.	$\sqrt{2gx}$

42. A wave represented by the equation $y = a\cos(kx + \omega t)$ is superimposed on another wave to form a stationary wave such that the point $x = 0$ is a node. The equation for the other wave is:
1. $a\sin(kx + \omega t)$
2. $-a\cos(kx + \omega t)$
3. $-a\cos(kx - \omega t)$
4. $-a\sin(kx - \omega t)$

43. Match **List-I** with **List-II**.

	List-I (x-y graphs)		List-II (Situations)
(a)		(i)	Total mechanical energy is conserved
(b)		(ii)	Bob of a pendulum is oscillating under negligible air friction
(c)		(iii)	Restoring force of a spring
(d)		(iv)	Bob of a pendulum is oscillating along with air friction

Choose the correct answer from the options given below:

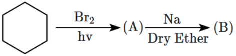
	(a)	(b)	(c)	(d)
1.	(iv)	(ii)	(iii)	(i)
2.	(iv)	(iii)	(ii)	(i)
3.	(i)	(iv)	(iii)	(ii)
4.	(iii)	(ii)	(i)	(iv)

44. Which, of the following equation represents a travelling wave?
1. $y = A\sin(15x - 2t)$
2. $y = Ae^{-x^2}(vt + \theta)$
3. $y = Ae^{x}\cos(\omega t - \theta)$
4. $y = A\sin x\cos\omega t$

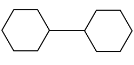
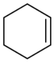
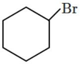
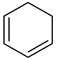
45. The frequency f and the length l of a simple pendulum are related by:
1. $fl =$ constant
2. $fl^2 =$ constant
3. $f^2 l =$ constant
4. $\frac{f^2}{l} =$ constant

Chemistry

46.



In the above reaction, product B is:

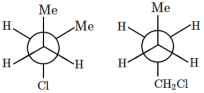
1.		2.	
3.		4.	

47. The structure of the product B formed in the reaction given below is:

$$CH_3 - C \equiv C - H \xrightarrow{(i) NaNH_2/NH_3(l)} (A) \xrightarrow{CH_3-CH_2-CH_2-Br} (B)$$

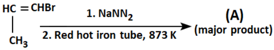
1. $CH_3 - C \equiv C - CH_2 - CH_2 - CH_3$.
2. $CH_3 - CH = CH_2$
3. $CH_3 - CH_2 - C \equiv C - CH_2 - CH_3$
4. $CH_3 - CH = C = CH - CH_2 - CH_3$

48. The given pair of structures represents:

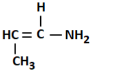
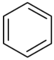
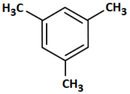


1.	Enantiomers	2.	Position isomers
3.	Conformers	4.	None of these

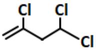
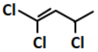
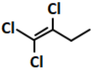
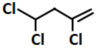
49. Given the following reaction:



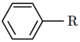
The major product A is:

1.	$CH_3CH_2CH_2NH_2$	2.	
3.		4.	

50. The structure of 1,1,3-trichlorobut-1-ene is:

1.		2.	
3.		4.	

51. Given the following benzyl/allyl system:

$R - C = CH_2$ (R is alkyl group) or 

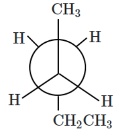
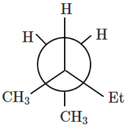
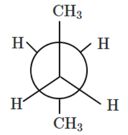
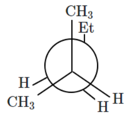
What is the correct decreasing order of the inductive effect?
1. $(CH_3)_3C- > (CH_3)_2CH- > CH_3CH_2-$
2. $CH_3CH_2- > (CH_3)_2CH- > (CH_3)_3C-$
3. $(CH_3)_2CH- > CH_3CH_2- > (CH_3)_3C-$
4. $(CH_3)_3C- > CH_3CH_2- > (CH_3)_2CH-$

52. Which of the following is the correct increasing order of reactivity of alkenes towards HX?
1. Isobutene > Propene > Ethene
2. Isobutene > Ethene > Propene
3. Ethene > Propene > Isobutene
4. Propene > Ethene > Isobutene

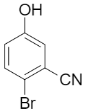
53. Which of the following meta-directing substituents in aromatic substitution is the most deactivating?
1. -C≡N
2. $-SO_3H$
3. -COOH
4. $-NO_2$

54. Which of the following compounds is the most reactive towards sulphonation?
1. Toluene
2. Chlorobenzene
3. Nitrobenzene
4. m-Xylene

55. Identify the conformer of 2-methylpentane:

1.		2.	
3.		4.	

56. The IUPAC name of the following compound is:



1. 4-Bromo-3-cynaophenol
2. 2-Bromo-5-hydroxybenzonitrile
3. 2-Cyano-4-hydroxybromobenzene
4. 6-Bromo-3-hydroxybenzonitrile

57. Incorrect statement regarding aromaticity among the following is:

1.	The compound should have a planar structure.
2.	The π–electrons of the compound are completely delocalized in the ring.
3.	The total number of π–electrons present in the ring should be equal to $(4n + 2)$ π electrons, where n = 0, 1, 2 … etc. This is known as Huckel's rule.
4.	The compound should have a linear structure.

58. Use the provided information in the following paper chromatogram.

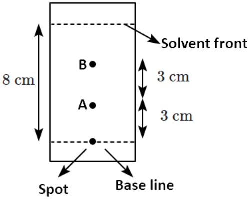


Figure: Paper chromatography for compounds A and B.

The R_f value of A is $Y \times 10^{-1}$. The value of Y will be:

1. 3.7
2. 2.6
3. 2.9
4. 1.0

59. Arrange the following hydrogen halides in order of their increasing reactivity with propene:

1.	HCl > HBr > HI	2.	HBr > HI > HCl
3.	HI > HBr > HCl	4.	HCl > HI > HBr

60. Match the following:

	Test/Method		Reagent
(i)	Lucas Test	(a)	$C_6H_5SO_2Cl$ / aq. KOH
(ii)	Dumas method	(b)	HNO_3 / $AgNO_3$
(iii)	Kjeldahl's method	(c)	CuO / CO_2
(iv)	Hinsberg Test	(d)	Conc. HCl and $ZnCl_2$
		(e)	H_2SO_4

1. (i)-(d). (ii)-(c), (iii)-(e), (iv)-(a)
2. (i)-(b), (ii)-(d), (iii)-(e), (iv)-(a)
3. (i)-(d), (ii)-(c), (iii)-(b), (iv)-(e)
4. (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d)

61. Given the following reaction:

$$CaC_2 \xrightarrow{H_2O} P \xrightarrow[\text{tube}]{\text{Hot iron}} Q \xrightarrow[\text{Anhydrous } AlCl_3]{CH_3Cl} R'$$

Product 'R' among the following is:

1. Benzene
2. Toluene
3. Aniline
4. Ethylbenzene

62. Heterolysis of propane gives:

1. Methyl and ethyl free radicals
2. Methylium cation and ethyl anion
3. Methyl anion and ethylium cation
4. Methylium radical and ethylium cation

63.

$$HC\equiv CH \xrightarrow[H_2O]{Hg^{2+}/H^+} A \xrightarrow{LiAlH_4} B \xrightarrow{P/Br_2} C$$

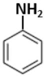
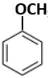
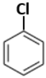
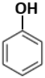
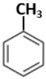
The final product C in the above sequence of reactions is:

1.	CH_3CH_2OH	2.	$CH_3CH_2 - Br$
3.	$BrCH_2CH_2Br$	4.	$\underset{Br}{H_2C} - \overset{H_2}{C} - OH$

64. Lassaigne's test for nitrogen is not given by:

1.	Nitrobenzene	2.	Hydroxyl Amine
3.	Aniline	4.	Pyridine

65. The correct order of reactivity of the following compounds towards electrophilic substitution reaction will be:

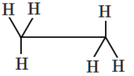
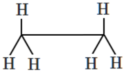
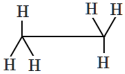
(I)		(II)	
(III)		(IV)	
(V)			

1. I > II > III > IV > V
2. I > II > IV > V > III
3. I > IV > II > V > III
4. I > II > IV > III > V

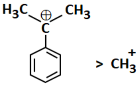
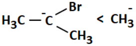
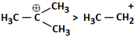
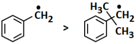
66. Which of the following statements about alkynes is incorrect?

1.	The first three members are gases.
2.	All alkynes are odourless.
3.	All alkynes are colourless.
4.	Alkynes are lighter than water and immiscible with water.

67. Which of the following representations of ethane depicts its non-staggered conformation?

1.		2.	
3.		4.	None of these

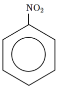
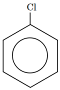
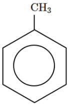
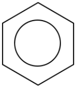
68. Match List-I with List-II and select the correct answer using the codes given below:

	List I (Stability)		List II (Reason)
(A)		(1)	Inductive effect
(B)		(2)	Resonance
(C)		(3)	Hyper-conjugation and resonance
(D)		(4)	Hyper-conjugation and inductive effect

Codes:
1. A - 2; B - 3; C - 4; D - 1
2. A - 3; B - 1; C - 4; D - 2
3. A - 4; B - 3; C - 1; D - 2
4. A - 3; B - 4; C - 2; D - 1

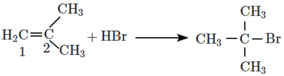
69. Which of the following sets includes nucleophilic species?
1. H^+, NH_3, Cl^-
2. BF_3, H_2O, NH_3
3. $AlCl_3, BF_3, NH_3$
4. $CN^-, H_2O, R-OH$

70. What is the correct order of the rate of the electrophilic aromatic substitution reaction (EASR) for the given compounds?

(a)		(b)	
(c)		(d)	

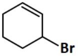
1.	c > b > a > d	2.	c > d > a > b
3.	a > b > c > d	4.	c > d > b > a

71. Under certain conditions, HBr adds to 2-methylpropene to give predominantly 2-bromo-2-methylpropane rather than isomeric 1-bromo-2-methylpropane. Which is the best explanation (mechanism wise) for this regioselectivity?

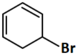


1.	H^+ adds preferentially to C-1 to form the more stable carbocation.
2.	Br^- adds preferentially to C-2 to form the more stable carbanion.
3.	$H^\bullet$ adds preferentially to C-1 to form the more stable radical.
4.	$Br^\bullet$ adds preferentially to C-2 to form the more stable radical.

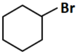
72. Arrange the following in decreasing order of stability of their transition state during elimination by a strong base.



I



II



III

1. II > I > III
2. II > III > I
3. I > III > II
4. I > II > III

73. Compounds with molecular formula $C_5H_{12}O$ cannot show :

1.	Tautomerism	2.	Position isomerism
3.	Metamerism	4.	Functional isomerism

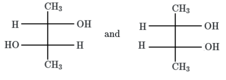
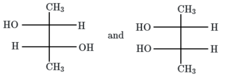
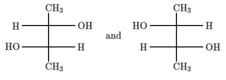
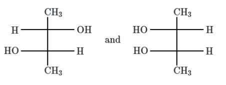
74. Consider the given two statements:

Assertion (A):	Propene reacts with HBr in the presence of H_2O_2 to give Markovnikov's addition product.
Reason (R):	H_2O_2 forms free radicals which cause homolytic cleavage of H-Br.

Select the correct option:

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

75. Which of the following pairs are enantiomers?

1.	
2.	
3.	
4.	

76. Match the items of Column I with items of Column II.

	Column I (Organic compound)		**Column II (IUPAC Name)**
a.	$(CH_3)_3\ CCH_2C(CH_3)_3$	i.	3,3-Di-tert-butyl-2,2,4,4-tetramethylpentane
b.	$(CH_3)_2\ C(C_2H_5)_2$	ii.	2,2,4,4-Tetramethylpentane
c.	Tetra – tert-butyl methane	iii.	3,3-Dimethylpentane

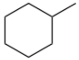
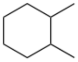
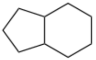
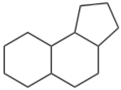
1.	a = i; b = ii; c = iii	2.	a = iii; b = ii; c = i
3.	a = ii; b = iii; c = i	4.	a = iii; b = i; c = ii

77. What is the impact of the nitro substituent on the reactivity and orientation of electrophilic substitution in nitrobenzene?

(a)	Deactivates the ring by inductive effect.
(b)	Activates the ring by inductive effect.
(c)	Decreases the charge density at the ortho and para positions of the ring relative to the meta position by resonance.
(d)	Increases the charge density at the meta position relative to the ortho and the para positions of the ring by resonance.

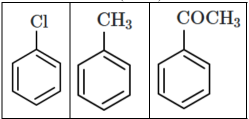
Choose the correct option:
1. **a** and **b**
2. **b** and **c**
3. **c** and **d**
4. **a** and **c**

78. The number of chiral carbons in each of the following compounds is respectively:

(I)		(II)	
(III)		(IV)	

1. 0, 2, 2, 4
2. 2, 2, 0, 4
3. 1, 2, 2, 4
4. 2, 2, 2, 4

79. The increasing order of the reactivity of the following compounds toward electrophilic aromatic substitution reactions (EASR) is:



(I)	(II)	(III)

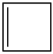
1. III < II < I
2. III < I < II
3. II < I < III
4. I < III < II

80. How many sigma (σ) and pi (π) bonds are present in the given structure?

$$(CH_3)_2CH-CH=CH-CH_2-CH=CH-\underset{\substack{|\\ C_2H_5}}{CH}-CH_3$$

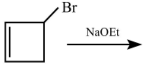
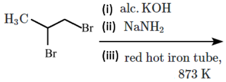
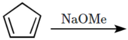
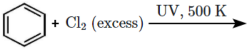
1. σ bonds -33 and π bonds -2
2. σ bonds -22 and π bonds -2
3. σ bonds -42 and π bonds -2
4. σ bonds -40 and π bonds -3

81. The most acidic hydrocarbon among the following is:

1.		2.	
3.		4.	

82. Geometrical isomers differ in:
1. Position of the functional group.
2. Position of atoms.
3. Spatial arrangement of atoms.
4. Length of the carbon chain.

83. Compounds that yield an aromatic compound as the major product are:

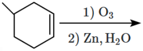
a.	
b.	
c.	
d.	

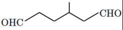
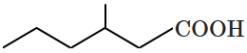
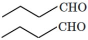
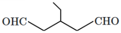
1. a, c
2. b, c
3. b, d
4. None of the above.

84. Why does the addition of HI in the presence of a peroxide catalyst not follow anti-Markovnikov's rule?

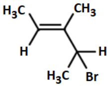
1.	HI is a strong reducing agent.
2.	The H–I bond is too strong to be broken homolytically.
3.	The I atom combines with H atom to give back HI.
4.	The iodine atom is not reactive enough to add across a double bond.

85. The major product of the following oxidation reaction is:



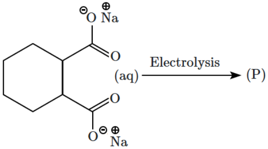
1.		2.	
3.		4.	

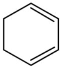
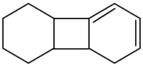
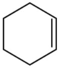
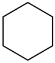
86. What is the IUPAC name of the following compound?



1. 3-Bromo-1, 2-dimethylbut-1-ene
2. 4-Bromo-3-methylpent-2-ene
3. 2-Bromo-3-methylpent-3-ene
4. 3-Bromo-3-methyl-1, 2-dimethylprop-1-ene

87. The major product (P), in the reaction given below is:



1.		2.	
3.		4.	

88. Which pair of groups exerts a $(-I)$ effect?
1. $-NO_2$ & $-CH_3$
2. $-NO_2$ & $-Cl$
3. $-Cl$ & $-CH_3$
4. $-CH_3$ & $-C_2H_5$

89. The electron pair in the $\pi-$ bond of an alkene has:

1.	33% p character and are at a lower energy level than the electron pair in the $\sigma-$ bond.
2.	50% p character and are at a higher energy level than the electron pair in the $\sigma-$ bond.
3.	100% p character and are at a lower energy level than the electron pair in the $\sigma-$ bond.
4.	100% p character and are at a higher energy level than the electron pair in the $\sigma-$ bond.

90. In CH_3CH_2OH, the bond that undergoes heterolytic cleavage most readily is:
1. $C-C$
2. $C-O$
3. $C-H$
4. $O-H$

Biology

91. Arrange the following events of meiosis in a correct sequence:
I. Crossing over
II. Synapsis
III. Terminalisation of chiasmata
IV. Disappearance of nucleolus

1.	II, I, IV, III	2.	II, I, III, IV
3.	I, II, III, IV	4.	II, III, IV, I

92. What did Schleiden and Schwann contribute to the cell theory?

1.	They discovered the nucleus.
2.	They proposed that all living organisms are made up of cells.
3.	They identified the cell membrane.
4.	They developed the first microscope.

93. What percentage of the cell cycle duration does the interphase last in a typical human cell with a 24-hour cycle?

1.	About 50%	2.	About 75%
3.	More than 95%	4.	Less than 50%

94. Adult human haemoglobin consists of:
I: 4 subunits.
II: 4 types of subunits.
1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

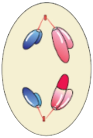
95. The cell shown in the diagram is most likely at:

1.	Metaphase I	2.	Metaphase II
3.	Anaphase I	4.	Anaphase II

96. Microtubules are the constituents of?

1.	spindle fibres, centrioles and cilia
2.	centrioles, spindle fibres and chromatin
3.	centrosome, nucleosome and centrioles
4.	cilia, flagella and peroxisomes

97. The cell shown in the given figure is most likely at:



1. Metaphase I
2. Metaphase II
3. Anaphase I
4. Anaphase II

98. The cofactor of the enzyme carboxypeptidase is:
1. Niacin
2. Flavin
3. Haem
4. Zinc

99. In plant cells:

I:	during the S phase, DNA replication begins in the nucleus.
II:	during the S phase, the centriole duplicates in the cytoplasm.
III:	during the G_2 phase, proteins are synthesised in preparation for mitosis
IV:	cell growth stops during S phase and G_2 phase.

1. **I** is correct; **II**, **III** & **IV** are incorrect
2. **I** & **IV** are correct; **II** & **III** are incorrect
3. **I** & **III** are correct; **II** and **IV** are incorrect
4. All the statements: **I**, **II**, **III** & **IV** are correct.

100. Which part of the chromosome holds the chromatids together?

1.	Kinetochore	2.	Telomere
3.	Chromatin	4.	Centromere

101. How are chloroplasts similar to mitochondria in terms of their internal composition?

1.	Both contain 80S ribosomes for protein synthesis.
2.	Both have inner membranes organized into stacks for metabolic processes.
3.	Both have double membranes, circular DNA, and 70S ribosomes.
4.	Both store pigments for trapping light energy.

102. Which phase of mitosis is marked by the initiation of condensation of chromosomal material?
1. Metaphase
2. Prophase
3. Anaphase
4. Telophase

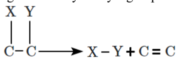
103. Nuclear pores:

I:	are formed by the fusion of the two nuclear membranes at some places.
II:	are the passages through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

104. A hypothetical double stranded DNA molecule contains 25 each of the four nitrogenous bases – A, T, G and C. The total number of hydrogen bonds in this molecule will be:
1. 50
2. 100
3. 125
4. 150

105. The type of biochemical reaction shown in the figure is catalysed by a group of enzymes called as:



1.	Transferases	2.	Lyases
3.	Ligases	4.	Oxidoreductases

106. Crossing over or recombination can be defined as:

1.	exchange of segments between sister chromatids of homologous chromosomes
2.	exchange of segments between non sister chromatids of homologous chromosomes
3.	exchange of segments between sister chromatids of heterologous chromosomes
4.	exchange of segments between non sister chromatids of heterologous chromosomes

107. Which of the following statements about mitochondria is true?

1.	Mitochondria contain their own DNA and ribosomes, which allows them to replicate independently within the cell.
2.	Mitochondria are primarily involved in the degradation of long-chain fatty acids.
3.	The outer membrane of mitochondria is highly permeable to protons, helping establish the proton gradient.
4.	Mitochondria are found only in animal cells and not in plant cells.

108. Consider the given statements regarding competitive enzyme inhibition:

Statement I:	Competitive inhibitor closely resembles the substrate in its molecular structure
Statement II:	In spite of its close structural similarity with the substrate, the inhibitor does not compete with the substrate for the substrate-binding site of the enzyme.
Statement III:	Inhibition of succinic dehydrogenase by succinate which closely resembles the substrate malonate in structure is an example of competitive inhibition.

1.	**Statement I** is correct; **Statement II** is correct; **Statement III** is correct
2.	**Statement I** is incorrect; **Statement II** is correct; **Statement III** is incorrect
3.	**Statement I** is correct; **Statement II** is incorrect; **Statement III** is incorrect
4.	**Statement I** is correct; **Statement II** is correct; **Statement III** is incorrect

109. The DNA content of a cell at G_1 phase is 2C. What will be the DNA content after S phase?
1. 2C
2. 4C
3. 6C
4. 8C

110. What can act as a competitive inhibitor of enzyme Succinic dehydrogenase?

1.	ATP	2.	$FADH_2$
3.	Malonate	4.	Pyruvate

111. Which of the following statements is incorrect about the primary structure of a protein?

1.	The primary structure determines the sequence of amino acids in a protein.
2.	The first amino acid in a protein is called the N-terminal amino acid.
3.	A protein exists as an extended rigid rod throughout its length.
4.	The last amino acid in a protein is called the C-terminal amino acid.

112. Which among the following scientists contributed the idea that all plants are composed of cells?
1. Matthias Schleiden
2. Rudolf Virchow
3. Theodor Schwann
4. Anton von Leeuwenhoek

113. Which of the following is an example of a prosthetic group?

1.	NAD+	2.	FAD
3.	Heme	4.	Coenzyme A

114. G_1 phase in a cell cycle:

I:	corresponds to the interval between initiation of mitosis and DNA replication.
II:	is characterised by a metabolically active cell that grows continuously and replicates its DNA.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

115. Crossing over:

I:	is the exchange of genetic material between two sister chromatids of heterologous chromosomes.
II:	is an enzyme-mediated process and the enzyme involved is called recombinase.
III:	leads to recombination of genetic material on the two chromosomes.
IV:	is completed by the end of pachytene, leaving the chromosomes linked at the sites of crossing over.

1. Only **I, II** and **III** are correct
2. Only **I, III** and **IV** are correct
3. Only **II, III** and **IV** are correct
4. **I, II, III** and **IV** are correct

116. Match each item in Column I with one in Column II and select the correct match from the codes given:

	Secondary metabolites		**Class**
A.	Vinblastine	P.	Polymeric substances
B.	Concanavalin A	Q.	Drugs
C.	Rubber	R.	Toxins
D.	Ricin	S.	Lectins

Codes:

	A	B	C	D
1.	R	S	P	Q
2.	Q	S	P	R
3.	Q	P	R	S
4.	P	Q	R	S

117. Identify the incorrectly matched pair:

	Structure	**Size range**
1.	Perinuclear space	10 – 50 nm
2.	Length of chloroplasts	2 – 5 μm
3.	Typical eukaryotic cell	10 – 20 μm
4.	Typical bacterium	1 – 2 μm

118. Match the following stages of mitosis with their key events:

	Stages of Mitosis		**Key Events**
(A)	Prophase	(i)	Chromosomes align at the equator
(B)	Metaphase	(ii)	Chromosomes decondense at poles
(C)	Anaphase	(iii)	Chromosomal material condenses
(D)	Telophase	(iv)	Centromeres split and chromatids separate

1. A-(ii), B-(i), C-(iii), D-(iv)
2. A-(iii), B-(i), C-(iv), D-(ii)
3. A-(i), B-(iii), C-(ii), D-(iv)
4. A-(iv), B-(ii), C-(i), D-(iii)

119. Which of the following statements is true about proteins?

1.	Proteins are homopolymers.
2.	Proteins are composed of repeating units of a single type of amino acid.
3.	Proteins are heteropolymers made up of 20 types of amino acids.
4.	Proteins are made up of only non-essential amino acids.

120. Consider the given statements about ribosomes:

I:	Ribosomes are non-membranous structures composed of rRNA and proteins.
II:	Eukaryotic ribosomes are smaller in size (70S) compared to prokaryotic ribosomes (80S).
III:	Ribosomes are involved in the process of protein synthesis.

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

121. Which biomolecule in living cells is neither a polymer nor a macromolecule?

1.	Lipid	2.	Carbohydrate
3.	Protein	4.	Nucleic acid

122. X-shaped structures, called chiasmata, are visible during:

1.	Zygotene of prophase I	2.	Pachytene of prophase I
3.	Diplotene of prophase I	4.	Diakinesis of prophase I

123. Meiosis involves:

1.	Two sequential cycles of DNA replication and nuclear division.
2.	One cycle of DNA replication and two cycles of nuclear division.
3.	Two cycles of DNA replication and one cycle of nuclear division.
4.	One cycle of DNA replication and one cycle of nuclear division.

124. Given below are two statements:

Statement I:	The Golgi cisternae are concentrically arranged near the nucleus with distinct convex *cis* or the forming face and concave *trans* or the maturing face.
Statement II:	The *cis* and *trans* faces of the organelle are identical and interconnected.

In the light of the above statements, choose the correct answer from the options given below:

1.	Both **Statement I** and **Statement II** are True.
2.	Both **Statement I** and **Statement II** are False.
3.	**Statement I** is True but **Statement II** is False.
4.	**Statement I** is False but **Statement II** is True.

125. Cells in the human body that secrete products like lipids and steroids are likely to have an abundance of:

1.	Smooth endoplasmic reticulum
2.	Rough endoplasmic reticulum
3.	Lysosomes
4.	Mitochondria

126. Which of the following enzyme classes is correctly matched with its function?

1.	Oxidoreductases: Catalyse the transfer of a group, other than hydrogen, between a pair of substrates
2.	Transferases: Catalyse the removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds
3.	Hydrolases: Catalyse the hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide, or P-N bonds
4.	Ligases: Catalyse the inter-conversion of optical, geometric, or positional isomers

127. Match each item in Column I with one in Column II and select the correct match from the codes given:

	Column I		Column II
A	Lysine	P	Simplest alpha amino acid
B	Serine	Q	Aromatic amino acid
C	Glycine	R	Basic amino acid
D	Tryptophan	S	R functional group is hydroxymethyl

Codes:

	A	B	C	D
1.	R	S	P	Q
2.	R	S	Q	P
3.	S	R	P	Q
4.	S	R	Q	P

128. An acrocentric chromosome is characterized by:

1.	A centromere located exactly in the middle, producing equal arms.
2.	A centromere at the extreme end, making it a telocentric chromosome.
3.	A centromere positioned near one end, creating a very short p arm.
4.	Multiple centromeres along its length.

129. Identify the correct statement regarding ribosomes:

1.	Ribosomes are found only in prokaryotic cells.
2.	Ribosomes are membrane-bound organelles in mitochondria and chloroplasts.
3.	Ribosomes are found in both eukaryotic and prokaryotic cells, as well as in mitochondria and chloroplasts.
4.	Ribosomes are not involved in protein synthesis in eukaryotic cells.

130. Which of the following polysaccharides is a homopolymer of fructose?

1.	Cellulose	2.	Glycogen
3.	Inulin	4.	Starch

131. Cells in our body, that are not dividing, are said to be at which phase of the cell cycle?

1.	G_1	2.	G_2
3.	G_0	4.	S phase

132. Which stage of prophase I of meiosis is characterised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers?

1.	Zygotene	2.	Pachytene
3.	Diplotene	4.	Diakinesis

133. Which German botanist examined a large number of plants and observed that all plants are composed of different kinds of cells which form the tissues of the plant?
1. Theodor Schwann
2. Rudolf Virchow
3. Jan Evangelista Purkyně
4. Matthias Schleiden

134. Chitin, a complex polysaccharide, is found in:

1.	Plant cell wall
2.	The exoskeleton of arthropods and cell walls of most fungi
3.	Bacterial cell wall
4.	Cartilage and other connective tissues

135. Consider the given two statements:

Statement I:	Coenzymes are organic molecules that assist enzymes in catalyzing reactions.
Statement II:	Coenzymes are permanently bound to the enzyme and are not altered during the reaction.

1. Both **Statements I** and **II** are correct.
2. **Statement I** is correct but **Statement II** is incorrect.
3. **Statement I** is incorrect but **Statement II** is correct.
4. Both **Statements I** and **II** are incorrect.

136. Which component of the endomembrane system of a eukaryotic cell is frequently observed in cells involved in protein synthesis and secretion?

1.	Rough endoplasmic reticulum
2.	Golgi complex
3.	Smooth endoplasmic reticulum
4.	Chloroplasts

137. A complex of ribosomes attached to a single strand of RNA is known as?
1. polymer
2. polypeptide
3. okazaki fragment
4. polysome

138. Some cells in adult animals do not appear to exhibit division. A typical example of such a cell will be:

1.	Skin epithelial cell	2.	Neuron
3.	Fibroblast	4.	Phagocytes

139. In some prokaryotes like cyanobacteria, there are membranous extensions into the cytoplasm which contain pigments and are called:
1. Mesosomes
2. Gas vacuoles
3. Inclusion bodies
4. Chromatophores

140. Consider the given two statements:

Statement I:	Mature sieve tube elements of vascular plants and mature human erythrocytes are not considered as 'living' cells.
Statement II:	Mature sieve tube elements of vascular plants and mature human erythrocytes lack a nucleus.

1. **Statement I** is correct; **Statement II** is correct
2. **Statement I** is incorrect; **Statement II** is incorrect
3. **Statement I** is incorrect; **Statement II** is correct
4. **Statement I** is correct; **Statement II** is incorrect

141. Consider the given two statements:

Assertion (A):	Competitive inhibitors reduce the rate of an enzyme-catalyzed reaction.
Reason (R):	Competitive inhibitors bind reversibly to the active site of the enzyme.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

142. Which of the following is a measure of the substrate concentration at which the enzyme catalyzes a reaction at half its maximum velocity?
1. K_m value
2. Pk_a value
3. Q10
4. ½ V_{max}

143. What describes a bivalent seen during meiosis-I?
1. Two chromatids and one centromere
2. Two chromatids and two centromere
3. Four chromatids and two centromere
4. Four chromatids and four centromere

144. Identify the correct statement with regard to G_1 phase (Gap 1) of interphase:

1.	The reorganisation of all cell components takes place.
2.	The cell is metabolically active and grows but does not replicate its DNA
3.	Nuclear division takes place
4.	DNA synthesis or replication takes place.

145. In a human cell cycle lasting 24 hours, approximately how much time does the interphase last?
1. 1 hour
2. 5% of the cycle
3. 95% of the cycle
4. 90 minutes

146. Synapsis occurs between:

1.	a male and a female gamete
2.	mRNA and ribosomes
3.	spindle fibres and centromere
4.	two homologous chromosomes

147. Match the following cellular processes with their corresponding terms:

	Column I (Process)		Column II (Term)
A.	Passive transport	1.	Energy-dependent movement of molecules
B.	Active transport	2.	Diffusion of water across a selectively permeable membrane
C.	Osmosis	3.	Movement of molecules along the concentration gradient
D.	Endocytosis	4.	Process of engulfing external materials into the cell

Options:
1. A-3, B-1, C-2, D-4
2. A-2, B-4, C-3, D-1
3. A-4, B-3, C-1, D-2
4. A-1, B-2, C-4, D-3

148. Cell membrane proteins can be classified as integral and peripheral on the basis of

1.	Their size	2.	Their function
3.	Ease of extraction	4.	Chemical nature

149. Consider the two statements:

Assertion (A):	Iodine-KI reagent can be used to detect starch in a sample.
Reason (R):	Starch can hold iodine molecules in its helical portion and starch-iodine is blue in colour.

1.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain**(A)**.
2.	**(A)** is True but **(R)** is False.
3.	Both **(A)** and **(R)** are False.
4.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.

150. What would not be a significance of mitosis?

1.	It ensures the growth and repair of tissues in multicellular organisms.
2.	It maintains the chromosome number constant across generations.
3.	It introduces genetic variation through crossing over and recombination.
4.	It is responsible for asexual reproduction in unicellular organisms.

151. The number of correct statements from the given statements regarding cell division in human somatic cells will be:

I:	The M Phase represents the phase when the actual cell division or mitosis occurs.
II:	The interphase represents the phase between two successive M phases.
III:	The interphase lasts more than 95% of the duration of cell cycle.
IV:	During interphase, the cell is metabolically not active.

1. 1
2. 2
3. 3
4. 4

152. In a polysaccharide, the individual monosaccharides are linked by a _________'A'________ bond, whereas in a polypeptide, amino acids are linked by a ____'B'_____ bond.
Select the option that correctly identifies 'A' and 'B':
1. A-Glycosidic, B-Peptide
2. A-Peptide, B-Ester
3. A-Amide, B-Hydrogen
4. A-Peptide, B-Glycosidic

153. Coenzymes differ from prosthetic groups because:

1.	Coenzymes are tightly bound to the enzyme, while prosthetic groups are transiently associated.
2.	Coenzymes are transiently associated, while prosthetic groups are tightly bound to the enzyme.
3.	Coenzymes are always composed of metal ions.
4.	Coenzymes are part of the enzyme's primary structure.

154. Which one of the following structures between two adjacent cells is an effective transport pathway?
1. Plasmodesmata
2. Plastoquinones
3. Endoplasmic reticulum
4. Plasmalemma

155. Consider the two statements:

Assertion (A):	Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms.
Reason (R):	Syngamy restores the diploid condition in sexually reproducing organisms.

1.	**(A)** is True; **(R)** is False
2.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**
3.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**
4.	**(A)** is False; **(R)** is True

156. The substrate has to go through a higher energy state or transition state during its conversion to a product in:

I:	Exothermic spontaneous reactions
II:	Endothermic reactions
III:	Enzyme catalyzed reactions

1. Only **I** and **II**
2. Only **I** and **III**
3. Only **II** and **III**
4. **I**, **II**, and **III**

157. Consider the two statements:

Assertion (A):	In a plant cell, the concentration of a number of ions is significantly higher in the vacuole than in the cytoplasm.
Reason (R):	The tonoplast facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

158. In meiosis, the splitting of the centromere of each chromosome takes place during:

1.	Metaphase I	2.	Metaphase II
3.	Anaphase I	4.	Anaphase II

159. Regarding polysaccharides:

I:	Cellulose does not contain complex helices.
II:	Inulin is a polymer of fructose.
III:	Left end of glycogen is the reducing end

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

160. Which of the following is not a basic shape of bacteria?
1. Bacillus (rod-like)
2. Coccus (spherical)
3. Helicoid (helix-shaped)
4. Spirillum (spiral)

161. Consider the given two statements:

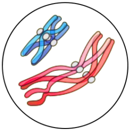
Statement I:	Enzymes are proteins with a specific three-dimensional structure that includes an active site.
Statement II:	Enzymes isolated from organisms living in hot environments are stable and retain their catalytic power even at temperatures up to 80°-90°C.

1.	**Statement I** is correct; **Statement II** is correct
2.	**Statement I** is correct; **Statement II** is incorrect
3.	**Statement I** is incorrect; **Statement II** is correct
4.	**Statement I** is incorrect; **Statement II** is incorrect

162. Certain prokaryotic cells possess gas vacuoles that aid in buoyancy. In which type of cells are gas vacuoles typically found?

1.	Animal cells
2.	Plant cells
3.	Blue-green and purple bacteria
4.	Fungal cells

163. The cell shown in the given figure is most likely at:



1. Prophase II
2. Prophase I
3. Anaphase I
4. Anaphase II

164. What is the typical width of the perinuclear space between the two membranes of the nuclear envelope?

1.	5 to 10 nm	2.	10 to 50 nm
3.	50 to 100 nm	4.	100 to 200 nm

165. Consider the given two statements:

Assertion(A):	The iodine test for starch is positive when the sample turns blue-black upon addition of iodine solution.
Reason(R):	Iodine molecules slip into the coil of the starch molecule, causing a color change due to the interaction between starch and iodine.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**
2.	Both **(A)** and **(R)** are True and **(R)** does not correctly explain **(A)**
3.	**(A)** is True, **(R)** is False
4.	**(A)** is False, **(R)** is False

166. A major site for synthesis of lipids is:
1. SER
2. Symplast
3. Nucleoplasm
4. RER

167. In a typical eukaryotic cell cycle, which phase is known as the resting phase where the cell is preparing for division by undergoing both cell growth and DNA replication?
1. M Phase
2. Interphase
3. G0 Phase
4. Cytokinesis

168. What is the inactive stage called where cells exit the G1 phase and do not divide further but remain metabolically active?

1.	G2 Phase	2.	S Phase
3.	M Phase	4.	G0 Phase

169. Which of the following will be applicable to the description of smooth endoplasmic reticulum?

a.	Frequently observed in the cells actively involved in protein synthesis and secretion
b.	Extensive and continuous with the outer membrane of the nucleus
c.	The major site for synthesis of lipid
d.	Synthesis of lipid-like steroidal hormones in animal cells

1.	a and b	2.	c and d
3.	a and c	4.	b and d

170. The correct description of prokaryotic ribosomes will be:

1.	They are about 15 nm by 20 nm in size and are made of two subunits - 50S and 30S units which when present together form 70S prokaryotic ribosomes.
2.	They are about 15 nm by 20 nm in size and are made of two subunits - 40S and 30S units which when present together form 70S prokaryotic ribosomes.
3.	They are about 20 nm by 25 nm in size and are made of two subunits - 60S and 40S units which when present together form 80S prokaryotic ribosomes.
4.	They are about 25 nm by 50 nm in size and are made of two subunits - 50S and 20S units which when present together form 70S prokaryotic ribosomes.

171. Which of the following correctly pairs an amino acid with its R group?
1. Glycine – Hydroxymethyl
2. Serine – Hydrogen
3. Valine – Aromatic group
4. Alanine – Methyl

172. A researcher is studying a plant species where the haploid number of chromosomes is 12 (n = 12). How many chromosomes would be visible in each daughter cell during metaphase of meiosis I?

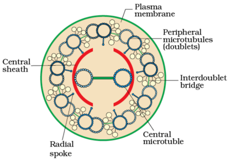
1.	24	2.	6
3.	12	4.	18

173. The number and linear sequence of amino acids in a protein are considered as the:
1. primary structure of a protein
2. secondary structure of a protein
3. tertiary structure of a protein
4. quaternary structure of a protein

174. A zwitterionic form of an amino acid exists when which of the following conditions are met in solution?

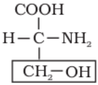
1.	The amino group is protonated, and the carboxyl group is deprotonated.
2.	Both the amino and carboxyl groups are neutral.
3.	Both the amino and carboxyl groups are protonated.
4.	Both the amino and carboxyl groups are deprotonated.

175. The given figure shows the ultrastructure of:



1.	Cilium	2.	Centriole
3.	Myosin	4.	Actin

176. Identify the amino acid:



1.	Glycine	2.	Serine
3.	Alanine	4.	Lysine

177. Lecithin, a small molecular weight organic compound found in living tissues, is an example of:
1. Phospholipids
2. Glycerides
3. Carbohydrates
4. Amino acids

178. During anaphase I of meiosis I:

1.	The homologous chromosomes separate, while sister chromatids remain associated at their centromeres.
2.	Simultaneous splitting of the centromere of each chromosome allows the sister chromatids to move towards opposite poles of the cell.
3.	The nucleolus disappears and the nuclear envelope also breaks down.
4.	the centrosome in animal cells undergoes replication.

179. Kinetochores:

I:	are small disc-shaped structures at the surface of the centromeres.
II:	serve as the sites of attachment of spindle fibres to the chromosomes.
III:	are same as centromeres.

1. **I** is correct; **II** & **III** are incorrect
2. **I** & **II** are correct; **III** is incorrect
3. **I**, **II** & **III** are correct
4. **I**, **II** & **III** are not correct.

180. Who first saw and described a live cell?
1. Robert Hooke
2. Anton Von Leeuwenhoek
3. Matthias Schleiden
4. Rudolf Virchow